

Figure 9.1: Different types of meristems on the basis of position in plant body

The term meristem was coined by **C. Nageli** 1858.

- The meristematic cells are isodiametric and they may be, oval, spherical or polygonal in shape.
- They generally have dense cytoplasm with prominent nucleus.
- Generally the vacuoles are either small or absent.
- Their cell wall is thin, elastic and made up of cellulose.
- These are most actively dividing cells.
- Meristematic cells are self-perpetuating.

Classification of Meristem

Meristem has been classified into several types on the basis of position, origin, function and division.

Theories of Meristem Organization and Function

Many anatomists illustrated the root and shoot apical meristems on the basis of number and arrangement and accordingly proposed the following theories – An extract of which is discussed below.

Shoot Apical Meristem

Apical Cell Theory

Apical cell theory is proposed by **Hofmeister** (1852) and supported by **Nageli** (1859). A single apical cell is the structural and functional unit.

Classification of Meristem

Position	Origin	Function	Plane of division
Apical meristem Present in apices of root and shoot. It is responsible for increase in the length of the plant, it is called as primary growth.	**Primary Meristem** It is derived from embryonic initials and differentiated into primary permanent tissues.	**Protoderm** It gives rise to epidermal tissue system and develops into epidermis,stomata and hairs.	**Mass meristem** It divides in all planes. Example: endosperm,young embryo and sporangium
Intercalary meristem Occurs between the permenant tissues. It is responsible for elongation of internodes.		**Procambium** It gives rise to primary vascular tissues. Example: xylem and phloem .	**Rib meristem or File meristem** It divides anticlinally in one plane. Example: development of cortex and pith
Lateral meristem Occurs along the longitudinal axis of stem and root. It is responsible for secondary tissues and thickening of stem and root. Example: vascular cambium and cork cambium.	**Secondary meristem** It is derived during later stage of development of the plant body. It produces cork cambium and interfascicular cambium.	**Ground Meristem** It gives rise to all tissues except epidermis and vascular strands.	**Plate meristem** It divides anticlinally in two planes. Example: development of epidermis

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